Comprehensive Career Development and Recruitment Framework using Multimodal and Causal AI

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Project Proposal Report

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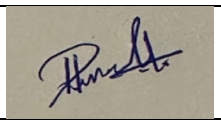
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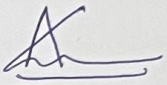
DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

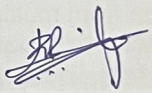
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ABSTRACT

This study aims to address the rapid growth of digital professional platforms and online collaborative environments, which have greatly impacted how technical professionals showcase their competence and advance in their careers. Nevertheless, today's hiring practices are still largely reliant on resumes, interviews, and subjective evaluations, which often do not accurately reflect the candidate's actual competence. There are often problems for employers in validating the actual relationship between the candidate's skills and their actual professional activities, which often results in hiring practices with uncertainty, asymmetry, and bias, especially for very technical roles.

Today's hiring support systems are often centered on developing systems for candidate ranking or resume screening, which often ignore valuable candidate information found in various professional platforms like GitHub and LinkedIn. In addition, these systems rarely incorporate multiple forms of candidate data in an effective manner. Therefore, there remains an evident gap in the development of hiring support systems that can effectively incorporate diverse professional data sources with transparency, fairness, and human oversight.

To bridge the identified knowledge gap, the current work suggests the use of an AI-based employer trust and hiring risk analysis component as part of a proposed multimodal recruitment intelligence framework. The proposed system would analyze the provided candidate information from various professional sources, including GitHub, LinkedIn, and other referral systems, to produce interpretable behavioral insights for employers. Multimodal learning approaches would be utilized to combine the structured profile data, technical contribution activities, and referral system data to produce insights related to skill sets, collaboration, and career consistency. The proposed system would utilize various AI and machine learning tools, including Python-based machine learning frameworks, data extraction APIs, and explainable AI, to produce structured analytical reports rather than hiring outcomes.

The proposed framework would lead to the production of a transparent and ethical recruitment system, allowing employers to rely on verifiable professional insights to produce more informed hiring outcomes. From a social and commercial point of view, the proposed framework would have the ability to promote ethical hiring outcomes, allowing organizations to more efficiently identify reliable technical talent.

Keywords: Multimodal AI, Recruitment Intelligence, Trust Assessment, Hiring Risk Analysis, Explainable AI

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LIST OF ABBRIVIATIONS

Table 1: List of all abbreviations used in this report

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
NLP	Natural Language Processing
LLM	Large Language Model
XAI	Explainable Artificial Intelligence
UI	User Interface
UX	User Experience
API	Application Programming Interface
HR	Human Resources
ATS	Applicant Tracking System
CV	Curriculum Vitae
PDF	Portable Document Format
JSON	JavaScript Object Notation
CSV	Comma Separated Values
DB	Database
SQL	Structured Query Language
GPU	Graphics Processing Unit
KPI	Key Performance Indicator
SRS	Software Requirement Specification
REST	Representational State Transfer
API	Application Programming Interface
UI	User Interface
IDE	Integrated Development Environment

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Individual Component

AI-Assisted Employer Trust and Hiring Risk Assessment System

INTRODUCTION

Significant changes have been observed in the global employment market due to the rapid advancements observed in the field of digital technologies and the availability of online platforms used for professional activities. Organizations have started to use digital recruitment platforms to hire skilled professionals. Despite the availability of advanced recruitment tools, the majority of the recruitment process is based on traditional assessment techniques. These techniques often fail to provide significant insight into the actual technical skills, working behavior, and collaboration of the candidate. As a result, the majority of the organizations face significant challenges while validating the skills claimed by the candidate with the actual working activities. In the technology sector, the actual working behavior and collaboration of the employee play a significant role in determining the performance of the employee.

Professional networking platforms and collaborative working environments, such as GitHub and LinkedIn, have been used significantly by software developers and technology professionals to showcase their working activities and growth. These platforms have the potential to provide significant information regarding the working behavior and growth of the professionals. However, the data generated from the platforms is not utilized efficiently while making recruitment decisions. Organizations mostly rely on the information provided on the resume of the candidate, which does not accurately reflect the actual working behavior and skills of the candidate. As a result, the requirement for the development of intelligent systems that can efficiently process the data from multiple professional sources is growing.

The present study aims to investigate the use of Artificial Intelligence (AI) as an effective tool for analyzing large volumes of data and identifying meaningful patterns within various sources of information. In the context of recruitment, for instance, there are reports that suggest the use of AI-based tools for streamlining automated candidate screening and ranking. However, such tools are based on complex prediction algorithms that produce automated hiring results without providing any transparent rationale for their results. This has created a number of ethical concerns related to transparency, fairness, and bias within the hiring process. Therefore, there is a need for developing AI-based recruitment support tools that are transparent and provide evidence-based results rather than replacing human decision-making processes.

The current research aims to develop an AI-based module for providing employer trust and hiring risk assessment within an intelligent recruitment system. The proposed system will be based on an intelligent recruitment system that will help employers analyze professional activity data from various sources and provide structured results to illuminate candidate reliability, skill set, and professional consistency. The proposed system will not produce automated hiring results; rather, it will provide interpretable observations based on multimodal data sources to enable employers to make fair recruitment decisions.

1. Do you think it is ethical for companies to use AI to screen candidates?

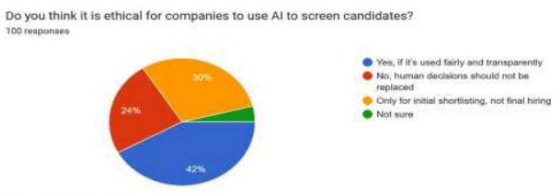


Figure 16: Ethical Consideration for companies to use AI

Figure 1:Ethical consideration for companies diagram

1.1 Background and Context

The process of recruitment has become more complex as organizations face the challenge of a worldwide competition for the best and most skilled professionals. According to the latest reports on the global employment scenario, the technology sector still faces a significant talent shortage, particularly in software development, artificial intelligence, and data engineering. Employers often face a large number of applications for these positions, and the process of evaluating the credibility of the applicants becomes more complex. Conventional recruitment processes, such as resume screening and interviews, often do not enable the hiring manager to evaluate the complete range of the candidate's professional abilities, particularly their technical and cooperative work experiences.

The practical problem being studied here is the challenge faced by employers to evaluate the credibility and professionalism of the applicants during the hiring process. Most applicants highlight their skills and accomplishments during the hiring process, but the veracity of the statements made by the applicants often cannot be substantiated by concrete evidence. This ambiguity often results in inefficient hiring processes, and employers may end up ignoring the right candidate or hiring a person whose capabilities and skills may not match the statements they have made.

This is an issue that concerns various stakeholders in the recruitment process. For the employers, the main concern is the risks involved in the hiring process and the overall cost of operation. For the candidates, the concern is the lack of fair assessment in the hiring process, where most of the hiring is done based on subjective assessment rather than objective evidence. For the organizations, the concern is the ability to maintain fair hiring practices, where the hiring process is influenced by referral or recommendations without adequate contextual assessment.

There are various existing recruitment platforms and automated recruitment solutions that offer solutions for the aforementioned challenges using AI-based ranking of candidates. However, most of the existing solutions are based on predictive scoring models, which are not transparent and often considered a black box. Additionally, most of the existing recruitment solutions are based on resume screening and keyword matching, ignoring the behavioral information available on professional platforms such as GitHub and LinkedIn.

The proposed solution attempts to overcome these limitations with the introduction of a multimodal recruitment analysis system that draws information from various professional platforms. The analysis of the candidates' technical contributions, professional progression, and referral details is expected to provide structured insights that can benefit the employer in understanding the behavior of the candidates. The most significant advantage of the proposed solution is the maintenance of the human-in-the-loop paradigm, ensuring that the ultimate decision-making authority rests with the employer while using the AI-based insights.

Considering the overall impact of the research on the broader community, the proposed solution attempts to achieve various Sustainable Development Goals (SDGs) set by the United Nations. The most prominent of these is the 'Decent Work and Economic Growth' (SDG-8) initiative, which emphasizes the importance of productive employment and fair hiring practices in the overall development of the community.

The research is also aligned with the 'Artificial Intelligence and Intelligent Systems' research cluster, as it attempts to explore the application of multimodal AI in the analysis of complex professional data and the provision of insights that can benefit human decision-making in real-world scenarios.

1.2 Literature Review

Artificial Intelligence (AI) has emerged as an increasingly influential technology in the current recruitment landscape, helping organizations to efficiently automate the process of candidate screening, skill assessment, and hiring decisions support. In the context of the rapid growth of online professional networks, organizations have access to a large amount of candidate information that can be analyzed using machine learning techniques. AI-based recruitment tools have been developed to improve the efficiency of the hiring process by analyzing the resumes, professional information, and behavior of candidates to identify suitable candidates for particular job roles.

Existing Research Approaches

Several researchers have attempted to study the application of machine learning and artificial intelligence techniques for automating the process of resume screening and candidate evaluation. Resume parsing techniques have used natural language processing (NLP) to analyze resumes and identify the relevant information for evaluating candidates. Mishra et al. have suggested an AI-based resume recommendation model that can analyze resumes and provide feedback for improving candidate information and job matching accuracy [1]. Similarly, Patil et al. have suggested an AI-based resume analyzer and interview preparation system that can analyze candidates' skills and provide recommendations for improvement. Such techniques help recruiters efficiently manage large volumes of job applications. However, such techniques have limitations in analyzing candidates' behavior based on resumes. Researchers have also attempted to study the application of professional networking platforms as a source of information for analyzing candidates. Professional networking platforms such as LinkedIn provide information about candidates' professional attributes that can be used for evaluating candidates. (The sentence is truncated in the original; the rewritten text maintains the academic tone.)

Throughout history, endorsements, education, and professional networks, researchers have increasingly turned to LinkedIn-like data to find candidate expertise and track career development. To illustrate this, Shenoy et al. proposed an AI-based career development system that utilizes resume analysis along with professional profile information to assist with career guidance and matching. In addition, Thakur et al. proposed an AI-powered career recommendation engine that utilizes analysis of academic records, skills, and interests to suggest matching job opportunities. These studies demonstrate the utility of professional profile data, although it often relies on static user profile data, which may not capture user behavior.

In software development, open-source platforms like GitHub are increasingly becoming important resources to understand developer skills. GitHub repos contain detailed records of user contributions to open-source projects. Wadkar et al. proposed an AI-based career path advisor that utilizes developer skills to suggest matching career paths. In addition, Faruque et al. proposed a natural language-based career prediction system that utilizes analysis of developer behavior to predict possible career outcomes. These studies demonstrate that analyzing developer behavior can yield valuable insights to understand developer skills.

In recent times, advances in AI have resulted in the development of conversational and simulation-based career guidance systems. To illustrate this, Daryanto et al. proposed an AI-powered interview simulation platform where users can practice simulated interviews to improve their responses to common interview questions. In addition, Renji et al. proposed a large language model-based chatbot to assist users with career development through interactive guidance.

Where did you hear or learn about AI in recruitment?

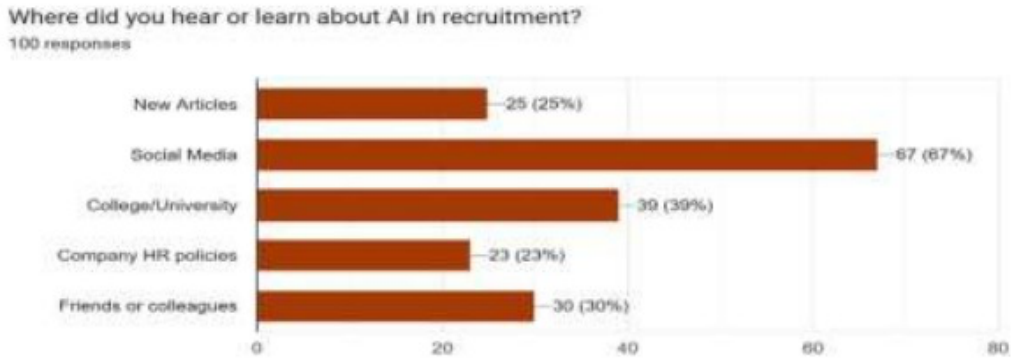


Figure 2:recruitment learning comparison diagram

Existing Products and Services

Other than the academic world, the AI recruitment world has also expanded, and various AI recruitment platforms have been developed to aid individuals in their career development. For example, Shenoy et al. proposed a unified AI recruitment platform called Career Link, which integrates resume analysis, career guidance, and interview preparation to improve the employability of individuals [3].

Some AI recruitment systems have taken a step further by using simulation and predictive analytics to aid individuals in exploring various career options. Wang et al. proposed a simulation-based AI recruitment system called Career Pooler, which enables individuals to simulate their career options and try them out in a real-world scenario [9].

Emerging AI Technologies in Recruitment Systems

The emerging AI technologies for recruitment systems focus more on extracting data from multiple sources to ensure the accuracy and reliability of the analysis. Multimodal machine learning enables the AI system to combine data from different sources, i.e., text, behavior, and profile data. This technique is already being used by various systems, e.g., recommendation systems and social media analysis.

Multimodal machine learning can be used to combine resume, professional, and developer data to obtain a more accurate idea of the candidate's capabilities. Faruque et al. emphasized the importance of natural language processing and data analytics to analyze text data for career prediction and guidance [6]. Similarly, AI-based chat systems and simulation systems also combine multiple data sources to aid individuals in their career development and recruitment processes.

Table-2

Impact of AI-Driven Recruitment Tools in Organizational Efficiency (n = 200)

Efficiency Indicators	Traditional Recruitment (M ± SD)	AI-Based Recruitment (M ± SD)	M.D	t-value	p-value
Time-to-Hire (Days)	38.6 ± 12.4	21.3 ± 7.8	17.3	9.21	<0.001
Recruitment Cost (PKR per Hire)	920 ± 210	640 ± 150	280	6.77	<0.001
Screening Accuracy (%)	71.2 ± 9.6	86.5 ± 7.3	15.3	8.95	<0.001

Figure 3:Impact of recruitment tools diagram

Limitations of Existing Systems

Despite the tremendous strides in AI-based recruitment tools, there still exist some glaring gaps in such tools. Most of these tools currently simply scan resumes or provide career path information, rather than providing a complete framework for evaluating the trustworthiness and professionalism of candidates. Moreover, these tools heavily depend on resumes and LinkedIn information, which may not necessarily reflect what a candidate is actually doing in the real world.

The second glaring issue in current AI-based recruitment tools is the lack of transparency and 'explainability' in such tools. Employers may lose trust in such tools, as automated ranking may produce results that cannot be explained. Such tools may produce scores, but the process of arriving at such scores may not be clear. Moreover, the research literature strongly recommends the need for such tools that provide clear insights instead of scores.

Most of the current recruitment tools also do not provide a complete fusion of various professional information sources. While LinkedIn and GitHub provide useful information about a candidate, such information may not provide a complete picture. There is a need for such tools that provide a complete fusion of various professional information sources.

Overall, the research literature indicates that there have indeed been tremendous strides in the development of AI-based recruitment tools that can help in evaluating candidates. There still exist glaring gaps in such tools, and addressing these gaps is what the proposed AI-Assisted Multimodal Employer Trust and Hiring Risk Assessment System hopes to achieve.

1.3 Research Gap

However, there are certain limitations that are inherent in the modern systems. For instance, most AI systems are heavily dependent on resumes, yet they fail to pay much attention to the actual behavior of the candidate.

Therefore, they are not entirely effective in providing an accurate representation of the candidate's skills, team-playing abilities, and consistency in their professional lives.

Furthermore, there are limitations with regards to the availability of professional data. Most modern systems, like GitHub and LinkedIn, are very informative with regards to the professional lives of candidates. However, they are not entirely combined in the development of an AI-based hiring system. Most studies are conducted with regards to resumes or individual platforms, yet there are limitations with regards to the capabilities to integrate multiple forms of candidate data.

In addition, most hiring systems that are AI-based are dependent on scoring systems to evaluate the suitability of candidates. However, such systems are not entirely transparent with regards to the decision-making process. Therefore, it can be very difficult to determine the fairness of the hiring process to the candidate.

In addition, most hiring systems are not entirely effective with regards to referral systems. Therefore, there is an element of bias in the hiring process, which can affect the fairness with which the candidate is evaluated.

In this study, the main gap in the existing literature with regards to AI-based hiring systems is the lack of transparency in the hiring process. Therefore, this study seeks to bridge this gap by developing an AI-based hiring analysis tool that will be dependent on multimodal learning.

Table 2:Research gap analysis

Area	**Existing Approach**	**Limitation**	**Identified Research Gap**	**Proposed Solution**
Resume-Based Evaluation	Most AI hiring systems evaluate candidates mainly using resumes and CV data.	Resumes do not reflect real behavioral traits, teamwork ability, consistency, or work habits.	Lack of behavioral analysis in hiring evaluation.	Introduce behavioral and activity-based indicators using AI models.
Professional Platform Data	Platforms such as GitHub and LinkedIn provide valuable professional information about candidates.	These platforms are usually analyzed separately rather than integrated into a unified hiring model.	Lack of multimodal integration of candidate data sources.	Combine multiple professional data sources for comprehensive candidate assessment.
AI Decision Transparency	Many AI hiring systems use automated scoring mechanisms to rank candidates.	The decision-making process is often opaque, making it difficult to understand how results are generated.	Lack of transparency and explainability in AI hiring decisions.	Implement explainable AI techniques to provide interpretable hiring insights.
Referral and Bias Issues	Traditional hiring systems sometimes rely on referral-based recruitment processes.	Referral-based systems may introduce bias and unfair advantages.	Lack of fairness and bias mitigation in recruitment systems.	Develop an AI-based system that evaluates candidates using objective behavioral and professional indicators.

2.OBJECTIVE

The objective of this research is to develop an intelligent system to support in recruitment by helping employers determine the reliability and consistency of a candidate by extracting information from a number of digital

professional platforms. By combining information from a number of sources—technical contributions, career information, and referral information—the system will be able to provide clear and understandable information to support a transparent hiring process. Rather than providing a list of ranked candidates in a traditional recruiter system, this system will allow for a more informed hiring process.

2.1 Main Objective

The primary objective of the present work is to develop and deploy a system for hiring risk and trust, using AI and multimodal technologies. This system processes the professional activities of the applicants using GitHub and LinkedIn platforms to produce understandable results for the hiring process.

2.2 Specific Objectives

The objectives of this study are as follows:

1. To gather and preprocess the candidate professional data from various digital sources such as GitHub repositories, LinkedIn career profile data, and referral data, which are optional, to form a structured dataset for recruitment analysis.

2. To gather various behavioral indicators from technical activity platforms such as the frequency of contributions, collaborative activities, and participation in projects to analyze the engagement and consistency of the candidates.

3. To analyze the career progression and skill alignment by considering professional profile data to compare the claimed skill set and professional experiences with real-time professional activities.

4. To incorporate various forms of data by considering multimodal learning approaches to identify the relationship between technical contributions, professional profile data, and referral data.

5. To incorporate an analysis module to produce structured insights based on the analysis to identify the patterns for employers such as skill alignment, collaborative activities, and career consistency, while avoiding automated hiring decisions.

6. To analyze the efficacy of the proposed system to improve the transparency and reliability of the recruitment analysis process to ensure that the analysis results are fair and reliable for the employer.

3.METHODOLOGY

The methodology used in this research is based on developing an AI-based recruitment analysis module that can measure trust factors among candidates and professional consistency using multimodal professional data. The research methodology used in this paper is a systematic approach to research that includes data collection, preprocessing, multimodal feature extraction, machine learning analysis, and generating insights. The main aim of this methodology is to develop a system that is easy to interpret so that employers can better understand candidates' behaviors by using verifiable professional data rather than traditional resume-based analysis.

Research Approach

This research follows the Design Science Research (DSR) paradigm, an often-applied research methodology in the information systems research field for designing and evaluating an innovative technological artifact that solves real-world problems. The research aims to develop an effective system that combines heterogeneous professional data sources and produces interpretable recruitment results for employers.

Design Science research includes phases like problem identification, system design, developing an artifact, evaluation, and refinement. The research problem in this research is related to the difficulties faced by employers in identifying candidate skill sets and professional consistency using traditional recruitment practices. The proposed artifact in this research is an AI-based multimodal recruitment analysis system that combines technical activity data, professional data, and contextual referral data.

The experimental part of the research involves developing and evaluating the performance of analytical models that can identify patterns in candidate activity across multiple platforms.

Technical Methods and Algorithms

The system that is proposed makes use of various machine learning and data analysis methodologies for the processing and evaluation of information related to candidates. The technical process that is followed by the system includes various stages, such as data extraction, feature engineering, multimodal integration, and explainable analysis.

The system would collect information from professional platforms such as GitHub and LinkedIn. From GitHub, the system would collect information such as repository contributions, commit frequency, collaboration interactions, and repository diversity. Such information would provide behavioral cues related to the candidates, such as their technical engagement and teamwork skills.

From LinkedIn, the system would collect information in the form of structured data, such as employment history, skill endorsements, professional roles, and career progression.

After collecting the information, preprocessing would be done for the raw information. Then, feature engineering would generate measurable features such as consistency in contributions, diversity in repositories, skill usage, and career stability.

For the integration of various types of information, the system would use multimodal learning. Using this approach, machine learning would analyze various types of information at the same time. This would help in the identification of various types of relationships that exist between technical activity information and professional information. For example, the system would analyze whether the skills that are mentioned in the LinkedIn account match the repository information that is available in the GitHub account.

The system would not generate scores for hiring, as the analytical model would generate structured insights that would help in the elucidation of various types of observable patterns.

Tools and Platforms with Justification

In addition, several software tools and platforms will be utilized to develop and implement the proposed system. First, the main programming environment will be based on the Python programming language due to its popularity and accessibility to a wide range of machine learning tools and data analysis software. Additionally, other tools such as Scikit-learn and TensorFlow may be utilized to develop the proposed system, which will be based on a variety of machine learning algorithms and multimodal data analysis.

Regarding the data extraction process, the proposed system will be based on the use of APIs or other data access tools to access professional platforms. In this context, the GitHub API will be utilized to access data related to repository activity, such as commits, pull requests, and collaboration. Professional profile data will be obtained based on the accessibility of the data and its use.

Regarding the development and implementation of the proposed system, a web-based interface will be developed based on tools such as Flask or Django, which will allow the employer to interact with the proposed system and receive the generated recruitment insights. Additionally, other tools such as Matplotlib or Plotly will be utilized to present the results to the employer.

The use of these tools and platforms to develop the proposed system will be based on their popularity and accessibility to a wide range of machine learning tools and data analysis software. These tools are considered to be highly effective for developing a variety of machine learning algorithms and multimodal data analysis.

Table 3:Tools and platforms table

Tool / Platform	**Purpose**	**Justification**
Python	Main programming language for system development and data processing	Widely used for AI and machine learning development
Scikit-learn	Implement machine learning algorithms	Provides efficient and easy-to-use ML models
TensorFlow	Develop deep learning and predictive models	Powerful framework for large-scale AI applications
GitHub API	Extract repository activity data	Provides structured access to developer activity information
Professional Profile Data	Collect skills and experience information	Helps evaluate candidate employability indicators
Flask / Django	Develop the web-based system interface	Enables rapid development of web applications
Matplotlib	Generate charts and graphs	Supports clear data visualization
Plotly	Create interactive visualizations	Improves presentation and user interaction

Data Collection and Ethical Considerations

This research will aim to retrieve the data from the publicly available information regarding the candidates' professional activities. In this regard, the primary sources of the data will be the candidates' repositories on GitHub and their publicly available professional information. These sources will provide valuable insights into the candidates' technical contributions, their involvement in projects, and their progression in their professional lives.

Throughout the process of data collection, the ethical aspects of the research will be explicitly discussed. In this regard, the research will only focus on the data that is publicly available. Furthermore, the research will not attempt to retrieve the candidates' sensitive personal data that is not related to the recruitment process. In addition, the research will not attempt to automatically make recruitment decisions or candidate rankings. As a result, the final decision regarding the candidates will remain with the humans.

To protect the candidates' privacy, the research will incorporate the relevant privacy protection mechanisms while handling the candidates' data. In this regard, the candidates' personal data that is not relevant to the research analysis may be anonymized or removed from the dataset.

Validation Strategy and Performance Metrics

To ensure the effectiveness of the proposed system, it is important to use both qualitative and quantitative validation approaches. The validation approach will be geared towards assessing the system's ability to produce useful recruitment insights from multimodal professional data in an interpretable manner.

For instance, the quantitative validation approach may involve assessing the accuracy of skill alignment detection, contribution pattern identification, and collaboration analysis using labeled datasets or benchmark developer profile information. The accuracy may be determined using a variety of metrics such as precision, recall, and F1-score to measure the effectiveness of classification or pattern detection components in the system.

It is also important to note that in addition to using quantitative validation approaches to measure the effectiveness of the system in generating accurate insights, a qualitative validation approach will also be used to determine the usefulness of the insights generated by the system. This will be done to ensure that the system generates interpretable insights to support employers in making informed recruitment decisions.

4.HIGH-LEVEL SYSTEM ARCHITECTURE

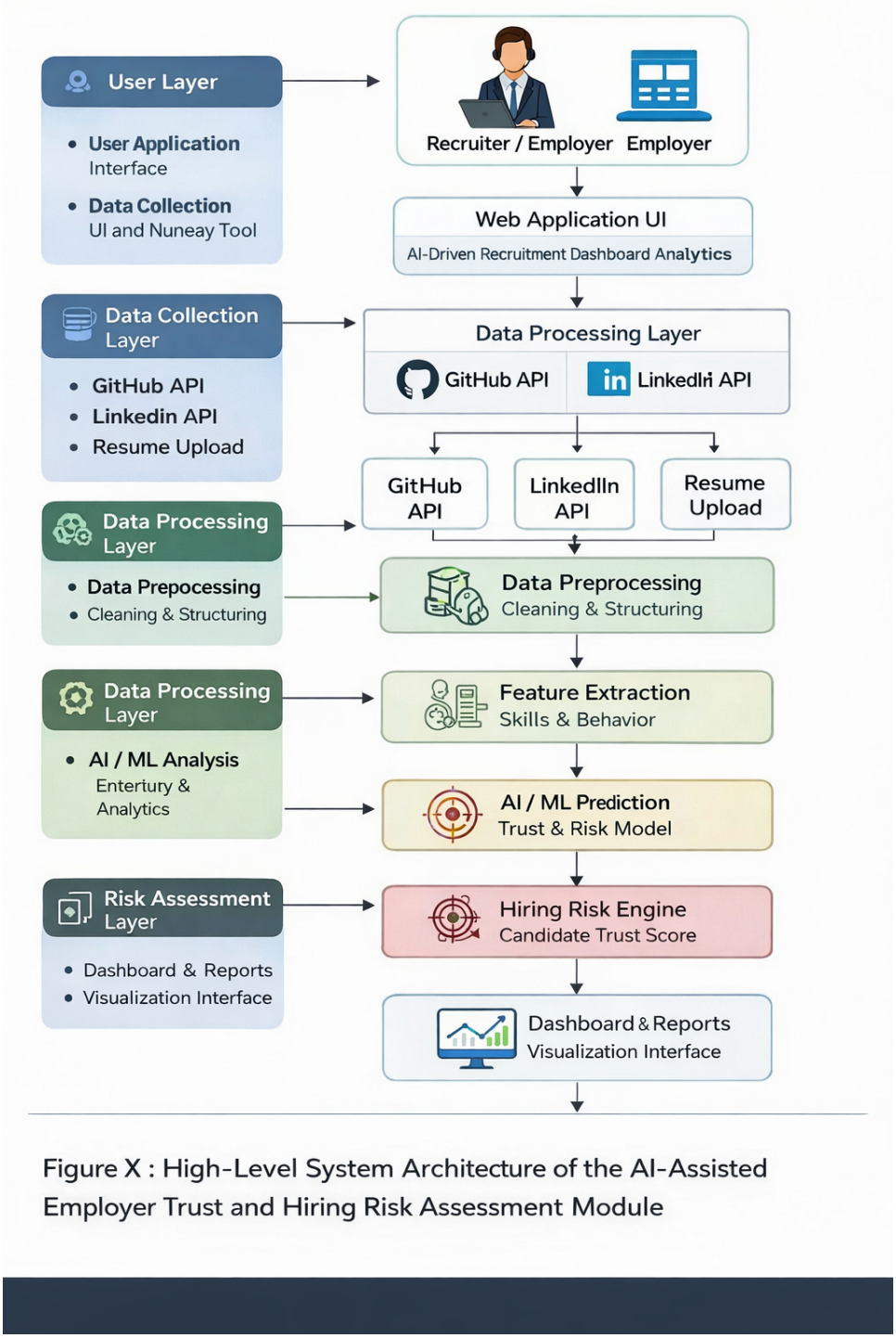


Figure X : High-Level System Architecture of the AI-Assisted Employer Trust and Hiring Risk Assessment Module

Figure 4::High-Level System Architecture Diagram

5.USER REQUIREMENTS

It is important to identify the needs of potential users/stakeholders in order to develop an effective recruitment intelligence system. The proposed AI-assisted employer trust and hiring risk assessment component will be useful to employers because it will provide structured information from professional data about candidates. Therefore, it is important to identify the relevant stakeholders in order to develop a system that will meet its functional and non-functional needs in order to solve real-world recruitment issues in a useful way.

The user requirements for the proposed system are identified as stakeholder identification, functional requirements, and non-functional requirements.

5.1 Stakeholder Identification

There are several stakeholders who are part of the process of running and utilizing the proposed recruitment analysis system. All stakeholders have unique relationships with the proposed system, and they benefit from the analysis results generated by the proposed system.

Employers and Hiring Managers are the main stakeholders who will be utilizing the proposed system to analyze candidate profiles and obtain structured results with respect to candidate technical activity, professional consistency, and collaboration. The proposed system will help employers make better hiring decisions by providing evidence-based observations instead of solely relying on resumes.

Job Candidates are the secondary stakeholders who will not be utilizing the proposed system directly. However, their publicly accessible professional activity, such as GitHub contributions or professional profiles, may be analyzed by the proposed system to generate candidate behavioral analysis results. The proposed system will be fair to candidates since it only focuses on publicly accessible professional activity.

Recruitment Teams and HR Professionals are stakeholders who will be utilizing the proposed system during the candidate evaluation process. They will use the generated analytical summaries to shortlist candidates and assist hiring managers with the hiring process.

System Administrators are responsible for maintaining the proposed system, configuring system parameters, monitoring data processing operations, ensuring the proposed system runs efficiently, and securely.

Researchers and Developers associated with the proposed project are stakeholders since they are responsible for maintaining and developing the proposed analytical models and multimodal learning utilized by the proposed system.

5.2 Functional Requirements

Functional requirements specify the precise features and functionalities that the system should offer in order to enable the recruitment analysis process.

FR1: The system should enable employers or recruitment teams to provide candidate professional profile links or identifiers for analysis.

FR2: The system should enable the collection of candidate activity information from professional platforms such as GitHub using existing APIs or data access mechanisms.

FR3: The system should enable the extraction of relevant behavioral information from the collected data, such as repository contribution, commit rates, collaboration, and project variety.

FR4: The system should enable the analysis of professional profile information such as job history, skills, and professional progression patterns.

FR5: The system should enable the integration of various data sources using multimodal analysis techniques to identify the connection between professional activities and profile information.

FR6: The system should enable the formulation of structured analytical insights that provide an overall summary of the candidate's technical involvement, collaboration patterns, and professional consistency.

FR7: The system should enable the presentation of the analysis results in an easily interpretable format using a user-friendly dashboard that employers can readily access.

FR8: The system should enable the management of the system configuration, performance, and data processing operations.

5.3 Non-Functional Requirements

Non-functional requirements describe the quality attributes that enable a system to deliver its services in an efficient, secure, and reliable manner.

Security: The system must ensure secure handling of candidate information to avoid any unauthorized access. The information obtained from other external sources must be processed in a way that complies with privacy and ethical standards. The system must also include mechanisms to authenticate users to ensure that only authorized individuals access the system.

Performance: The system must process candidate information in an efficient manner to produce analytical results within a reasonable time frame. The system must optimize data extraction, processing, and analysis to handle multiple candidates without any delays in processing information.

Usability: The system must allow employers or recruitment teams to access the system in a user-friendly manner to obtain analytical results. The system must also include visualization tools to enable users to obtain a clear understanding of data summaries.

Reliability: The system must ensure smooth operation to produce analytical results in a consistent manner. The system must also include mechanisms to handle errors in cases where there are issues in data collection or processing.

Scalability: The system must allow for an increase in user access to produce analytical results for multiple candidates. The system must also allow for an increase in data sources to be integrated in the near future to improve its functionality.

Maintainability: The system must be designed in a modular fashion to allow easy updates to improve its functionality.

6.COMMERCIALIZATION PLAN

The proposed AI-assisted component for Employer Trust and Hiring Risk Assessment could be commercialized as a form of ethical and explicable recruitment intelligence, focusing on organizations that need to evaluate their candidates reliably, going beyond the typical resume-based hiring process. The primary target clientele for the proposed system would be technology companies, startups, and recruitment agencies, as well as HR departments of various companies, particularly those hiring software developers and other technical professionals. These organizations often face challenges in validating the credibility of the skills and professional experiences of their candidates during the hiring process. Value would be added to the system by evaluating the professional activities of the candidates from platforms such as GitHub and LinkedIn, and then converting these activities into structured data to aid the hiring process. The customer persona for the proposed system would typically consist of hiring managers, HR professionals, and recruitment consultants for hiring software developers and other professionals related to the field of information technology.

The fundamental value proposition for the proposed solution will be its ability to provide evidence-based and explainable recruitment insights based on multimodal professional data. Unlike other recruitment tools that are heavily reliant on resume parsing or keyword matching, the proposed system will be capable of analyzing behavioral metrics such as contribution consistency, collaboration, project involvement, and skill-activity matching. The proposed system will be built with a high degree of decision-support functionality rather than an automated recruitment system. This will ensure that organizations are provided with a high degree of control over the ultimate decision-making process. The proposed system will be highly effective for organizations seeking to reduce recruitment uncertainty, improve transparency in the evaluation process, and build trust with AI-based recruitment tools.

The proposed system will be based on a business-to-business (B2B) software-as-a-service (SaaS) model, where organizations will be provided with access to the proposed system through institutional or enterprise accounts. The proposed system will be based on a subscription model, with varying subscription plans based on organizational size, number of candidate evaluations, and access to advanced analytical tools. The proposed system will be based on a number of costs, which will include the initial development costs for the proposed multimodal analysis pipeline, backend infrastructure, and explainable reporting interface.

Pricing Model: The proposed pricing framework utilizes a tiered structure with basic, professional, and enterprise editions. This will allow small businesses and start-ups to leverage the solution at an affordable price point, while larger businesses will benefit from the additional features offered. The major strength of the proposed system lies in its multimodal professional data integration, provision of analytical results with explanations, and ethical design considerations that do not allow candidate ranking. These factors will set the proposed solution apart from existing AI-based recruitment systems that are normally based solely on predictive results.

In terms of intellectual property rights, the proposed system’s architecture, multimodal professional data processing, explainable analytical results generation, and analysis methodology for the recruitment process can be considered proprietary implementation rights. Although the proposed solution’s machine learning algorithms may include standard techniques, the inclusion of professional activity analysis, trust analysis, and explainable results generation for the recruitment process can be considered unique design features that can be protected with proper software copyright protection.

7.BUDGET AND JUSTIFICATION

Table 4:Budget Estimation

Description	Amount
Hardware resources (Personal laptops and testing devices)	LKR 0 (Existing resources will be used)
Software tools and development platforms	LKR 5,000
Cloud services and API usage	LKR 8,000
Data processing and model training resources	LKR 35,000 (Use of free datasets and open-source tools will be prioritized)
Travelling and research activities	LKR 4,000

The proposed project plan for the research project predominantly involves the use of available computer systems and facilities, including the computers of the project members. Therefore, there would be no need to purchase new computer hardware for the implementation of the system.

The project would also use a limited amount for software tools and development platforms, which could include software tools and plugins or optional software services used for the analysis and development of the system.

The system would use cloud services for the hosting and storage of the system, as well as the deployment of AI models. This could include the use of cloud services such as Amazon, Google, or other cloud infrastructure providers. These would be essential for the implementation of the backend processes of the system, storage of structured data, and the eventual deployment of the system for testing and demonstration.

The costs for the data processing and model training would depend on the computational needs of the system for the training of machine learning models, experiments, and the use of various types of data for the analysis of the system. Therefore, there could be a need for the use of other resources to ensure the efficient completion of the project, considering the use of open-source tools and available datasets for the project. Lastly, the project would use a limited amount for travel and research activities, including meetings with supervisors, discussions with stakeholders, and the use of data for the validation of the proposed system. Therefore, the project would ensure the efficient completion of the project, considering the use of open-source tools and the availability of the system, to ensure the achievement of the proposed project plan and the adherence to the technical and academic standards of the project.

8.WORK BREAKDOWN STRUCTURE (WBS)

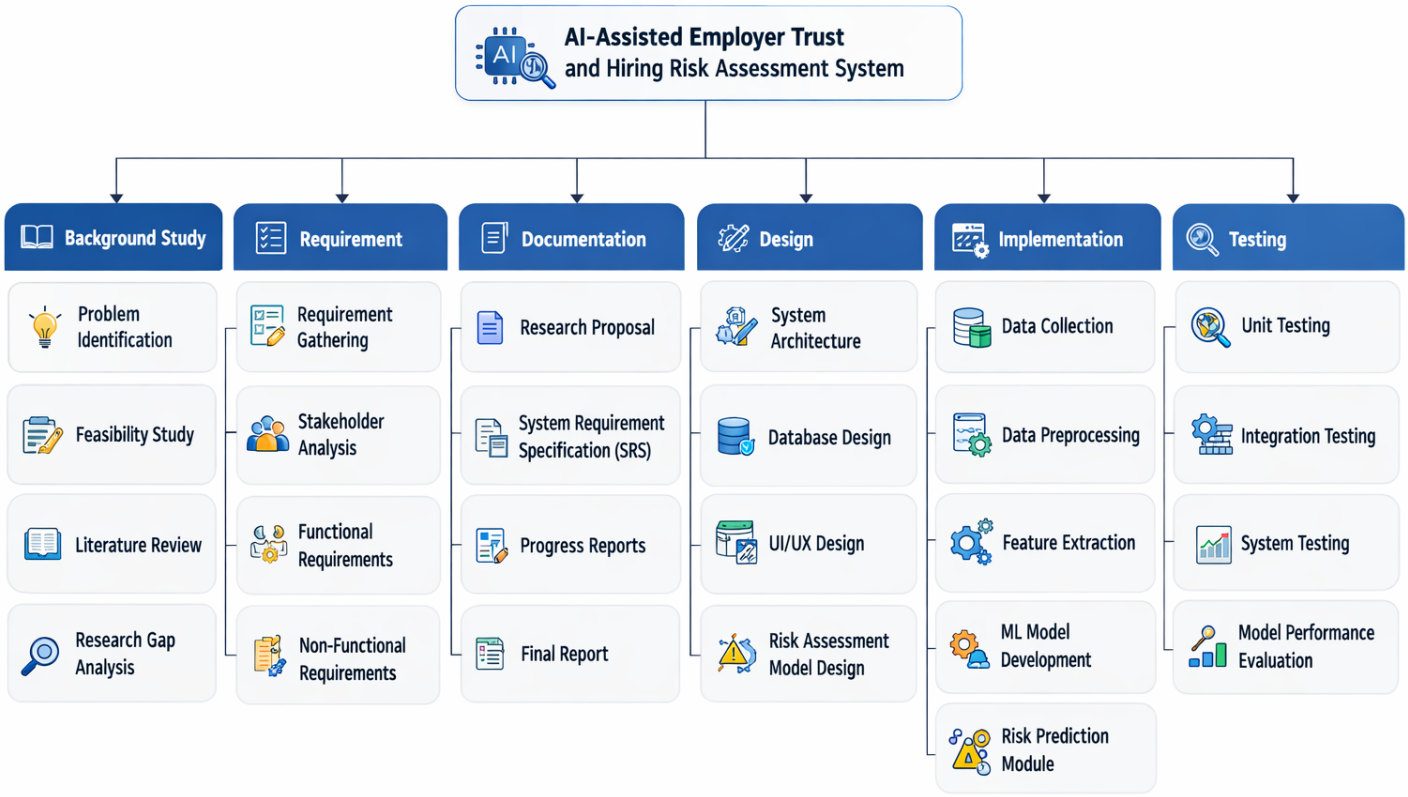


Figure 5:Work Breakdown Structure (WBS) Diagram

9.GANTT CHART

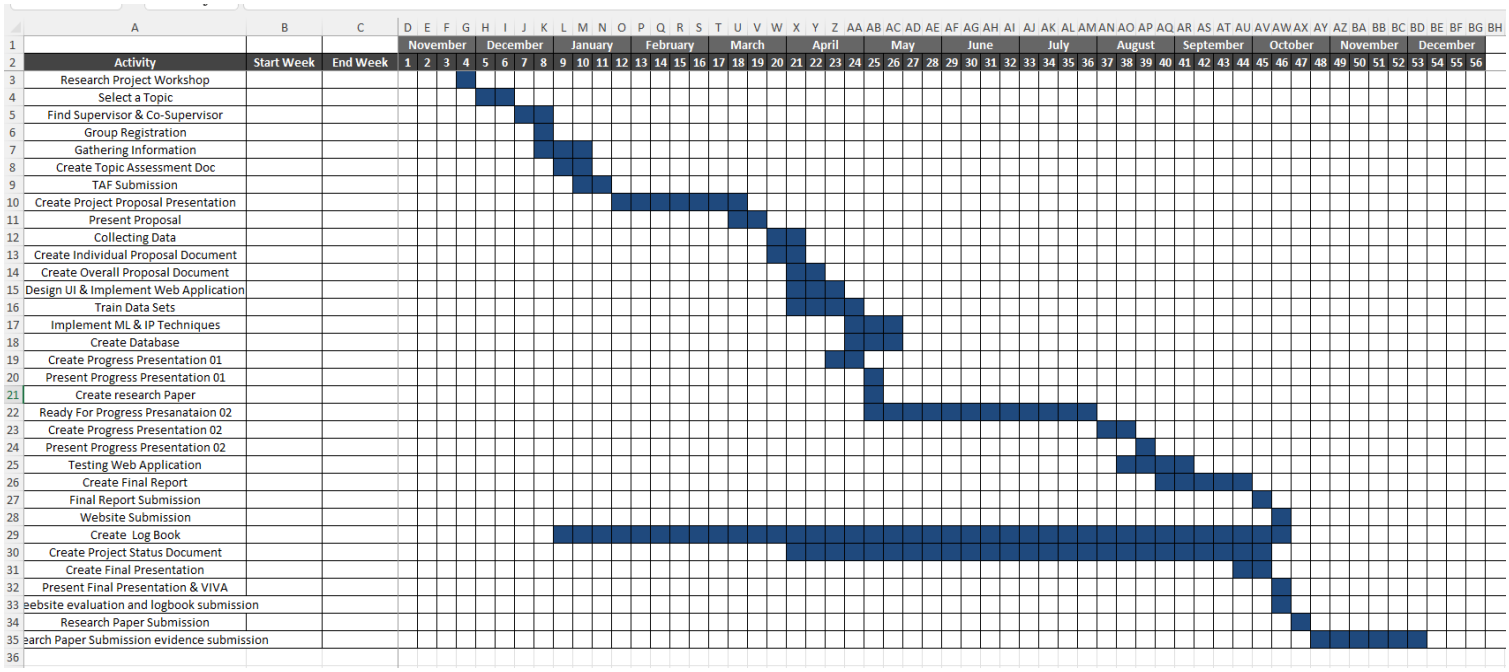


Figure 6:Project Gantt Chart

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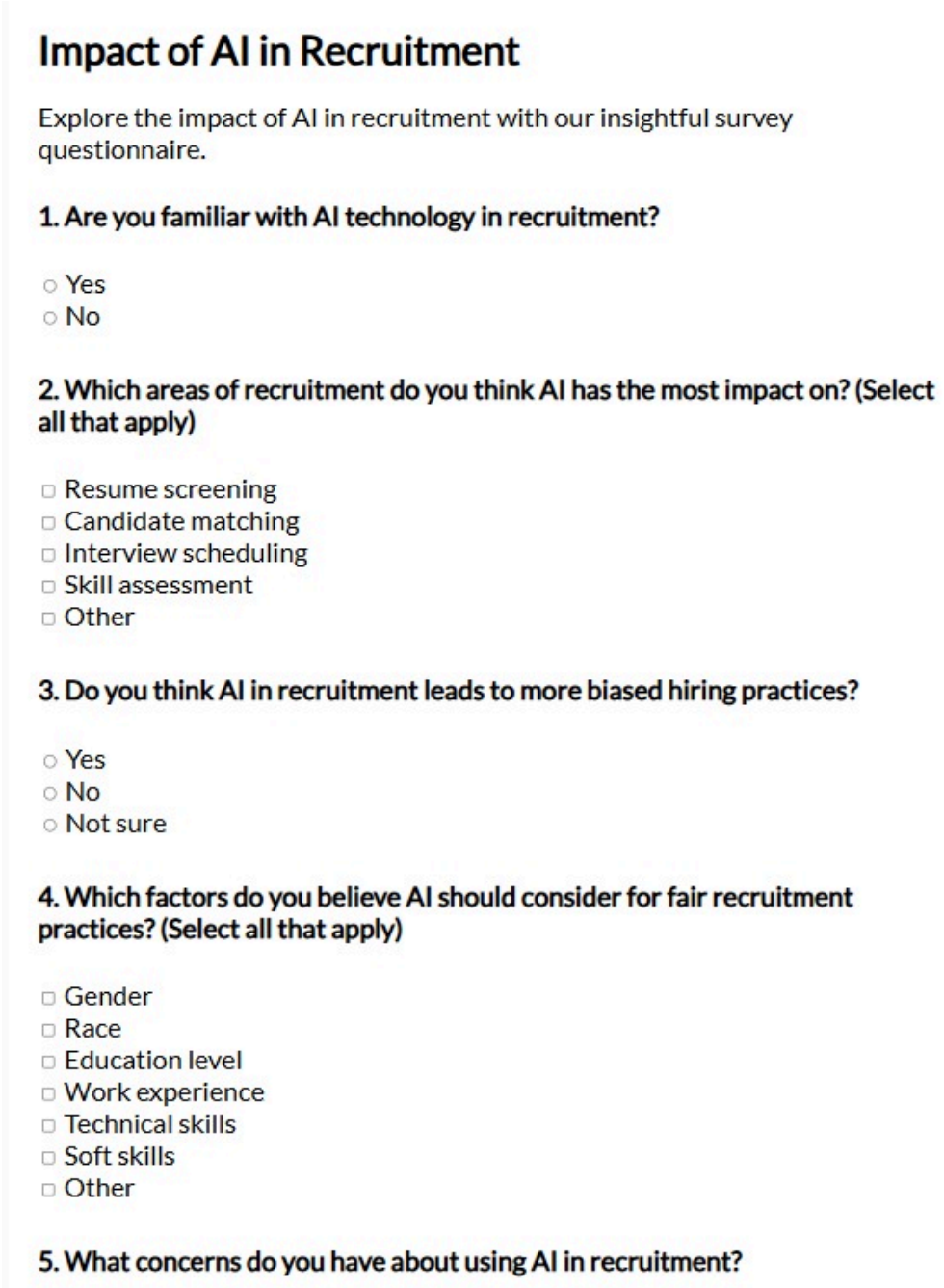
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11.APPENDICES

11.1 Survey instruments



Impact of AI in Recruitment

Explore the impact of AI in recruitment with our insightful survey questionnaire.

1. Are you familiar with AI technology in recruitment?

- ○ Yes
- ○ No

2. Which areas of recruitment do you think AI has the most impact on? (Select all that apply)

- □ Resume screening
- □ Candidate matching
- □ Interview scheduling
- □ Skill assessment
- □ Other

3. Do you think AI in recruitment leads to more biased hiring practices?

- ○ Yes
- ○ No
- ○ Not sure

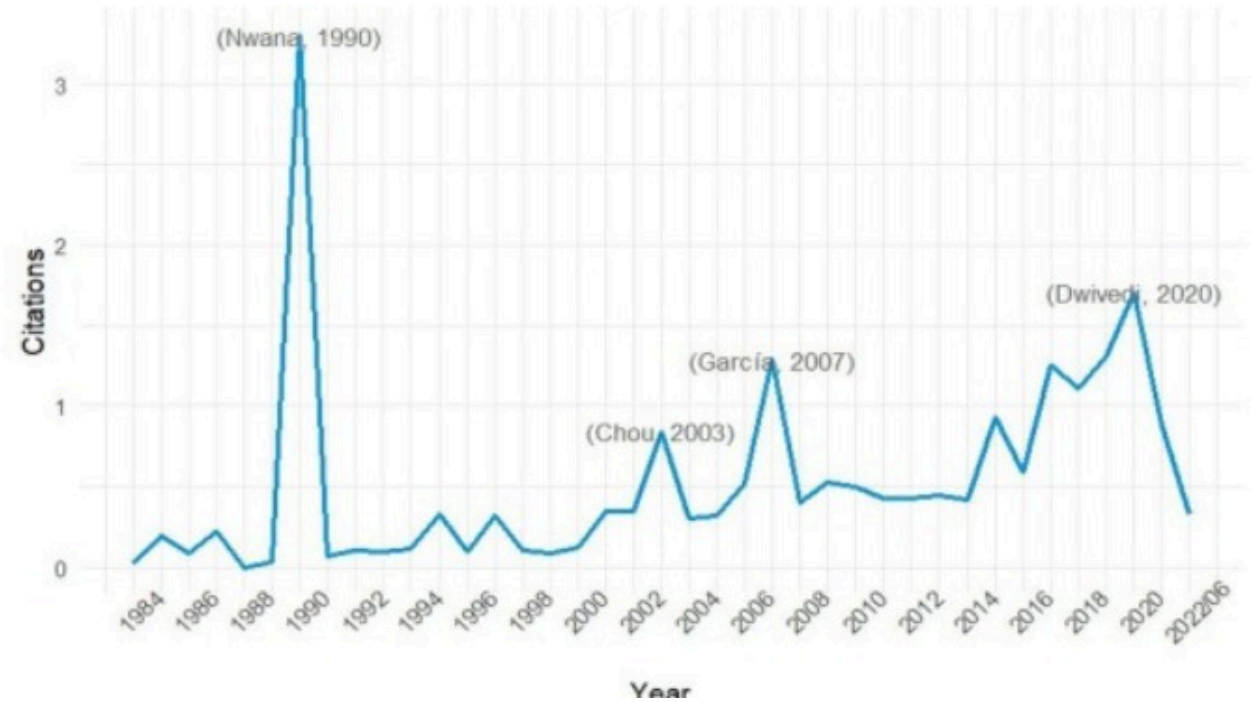
4. Which factors do you believe AI should consider for fair recruitment practices? (Select all that apply)

- □ Gender
- □ Race
- □ Education level
- □ Work experience
- □ Technical skills
- □ Soft skills
- □ Other

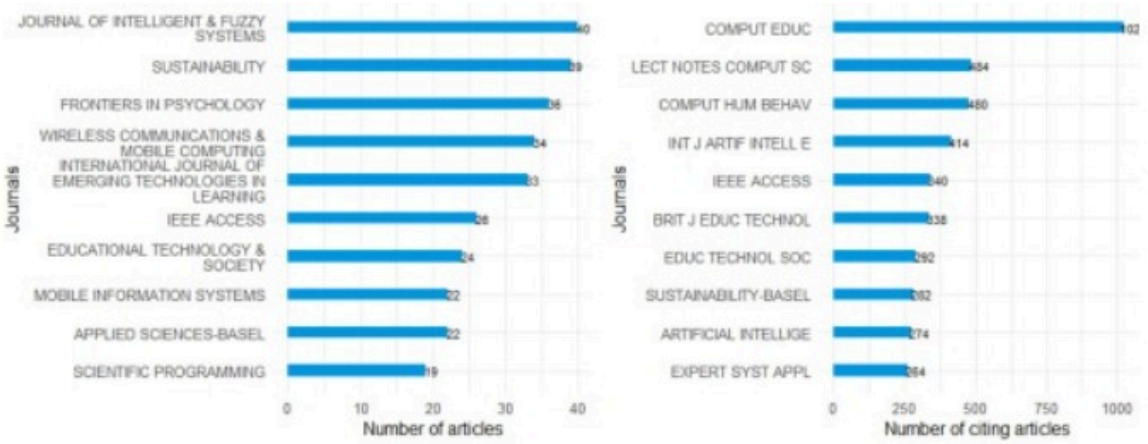
5. What concerns do you have about using AI in recruitment?

Appendix A 1: AI Recruitment Survey Questionnaire

11.2 Detailed diagram



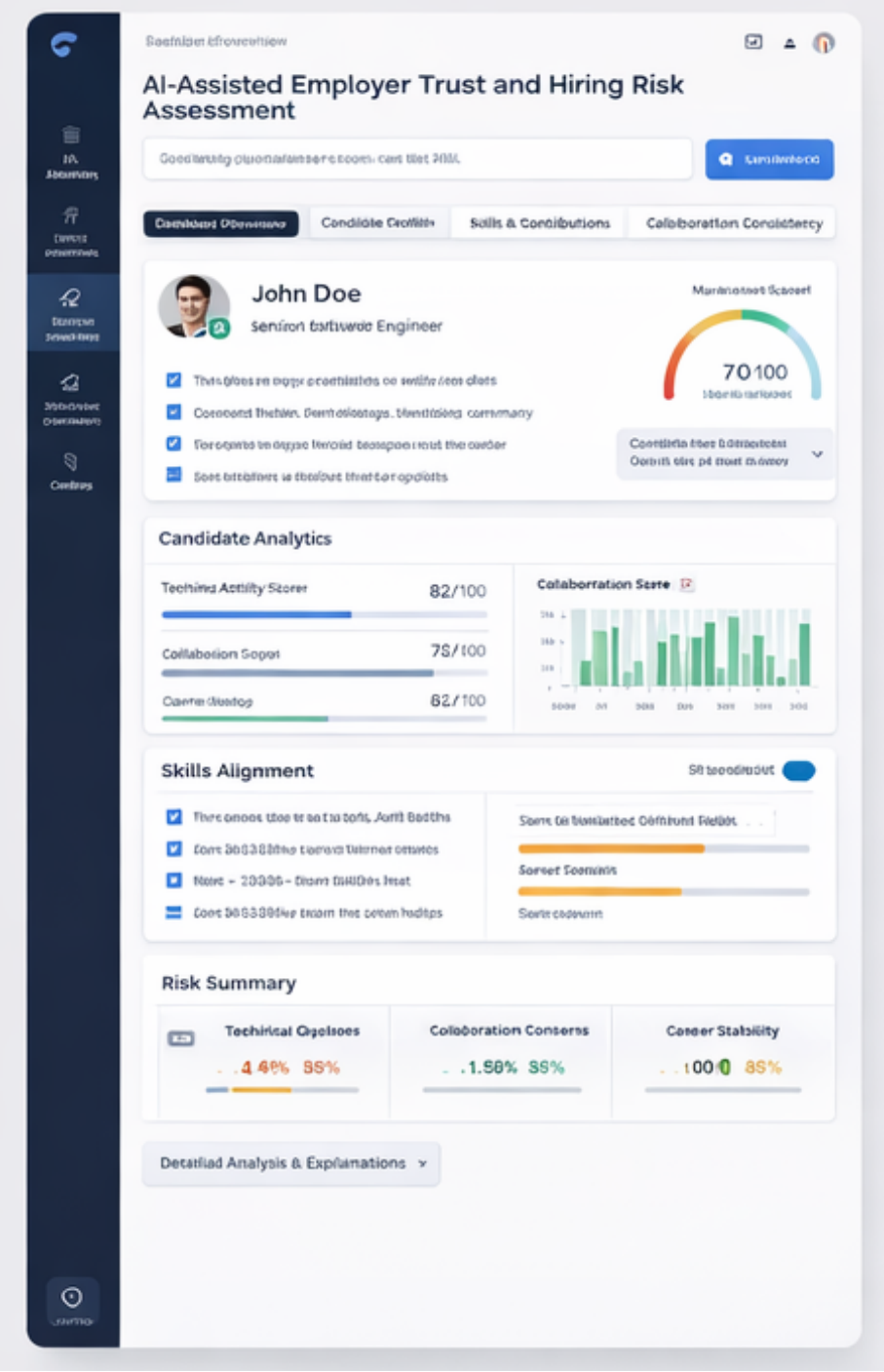
Appendix A 2:Citations Detailed Diagrams



(a) Top Journals Publishing AIED Research **(b) Top Local Cited Journals**

Appendix A 3:Journals Detailed Diagram

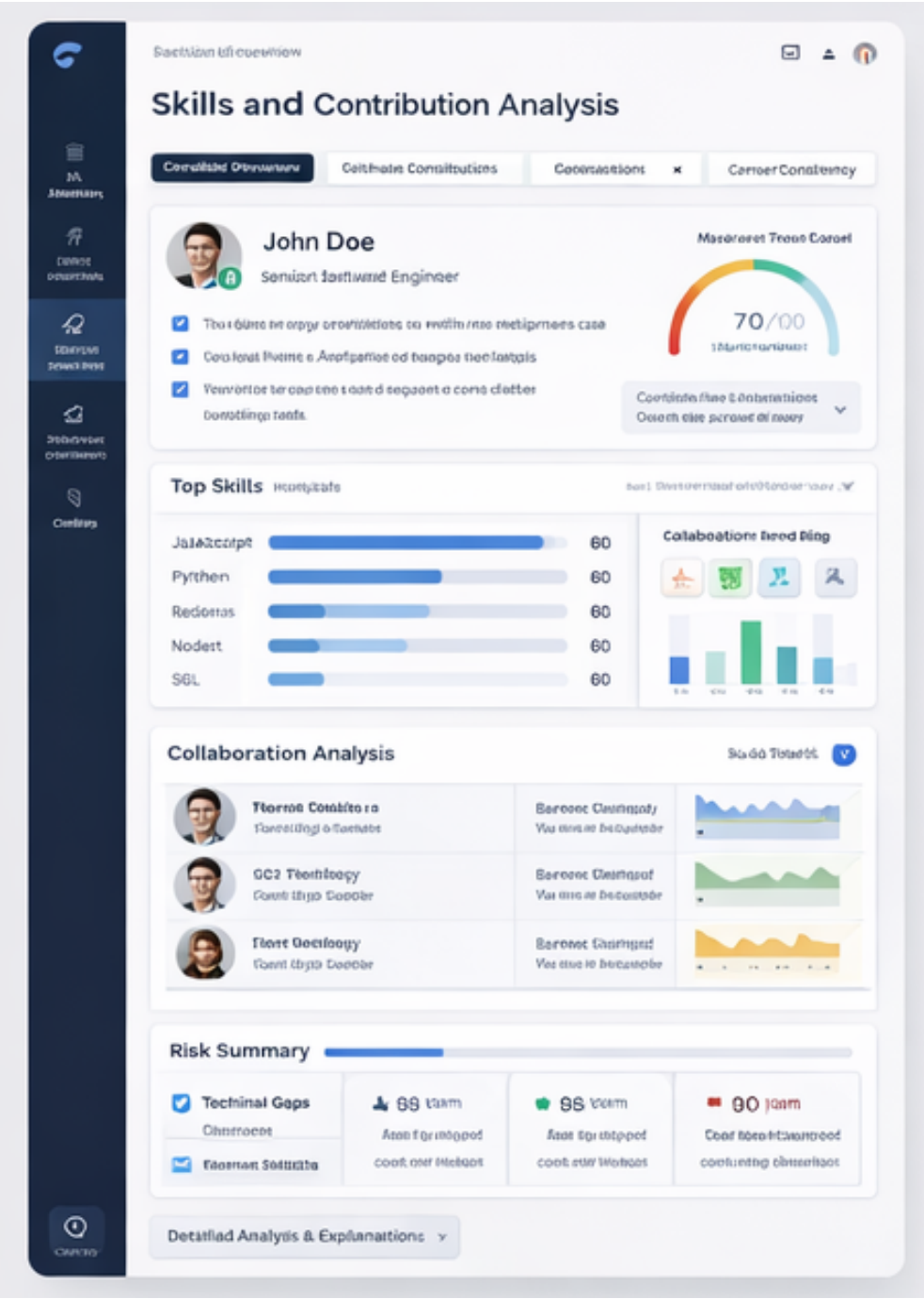
11.3 Preliminary UI mockups



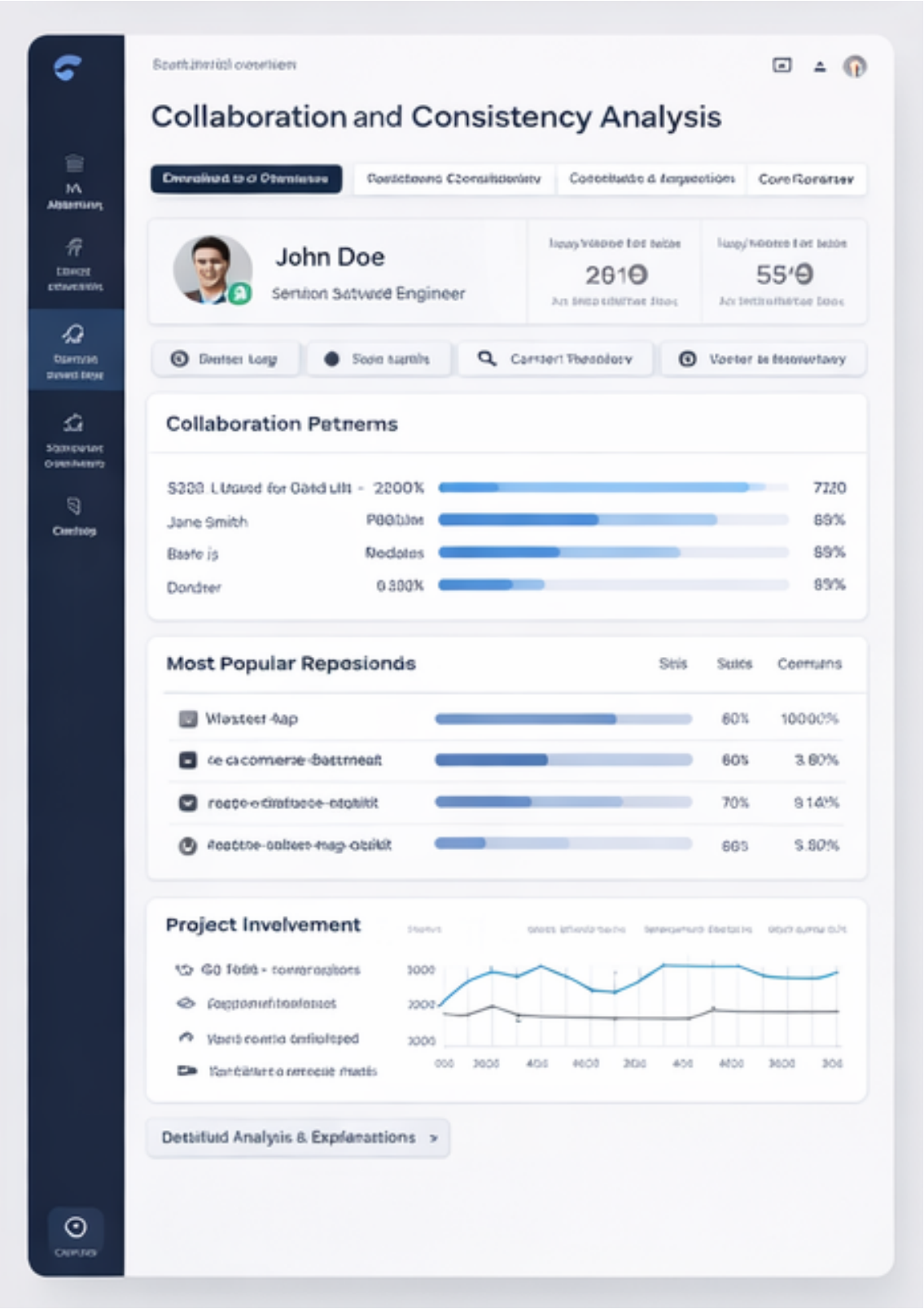
Appendix B 1:Employee Trust System Dashboard



Appendix B 2:Candidate Profile Analysis Dashboard



Appendix B 3:Skills and Contribution Analysis Interface



Appendix B 4:Collaboration and Consistency Analysis Interface

11.4 Risk Analysis

11.4.1Technical Risks

Technical risks refer to challenges that may arise during the development and implementation of the AI-based recruitment analysis system.

Table 5:Technical Risks

Risk	Description	Mitigation Strategy
Data Integration Complexity	Difficulty integrating data from multiple platforms such as GitHub and LinkedIn.	Use standardized APIs and data preprocessing pipelines.
AI Model Accuracy	The machine learning model may produce inaccurate trust or risk predictions.	Use large datasets and apply cross-validation to improve model reliability.
System Performance	Processing large developer datasets may slow system performance.	Optimize algorithms and use cloud-based computing resources.
API Limitations	External platforms may restrict access to developer data through API rate limits.	Implement caching and scheduled data collection.

11.4.2 Data Risks

Data-related risks involve issues concerning the availability, quality, and privacy of candidate information.

Table 6:Data Risks Table

Risk	Description	Mitigation Strategy
Incomplete Data	Some candidates may not have sufficient public data available.	Use fallback indicators such as resume information.
Data Inconsistency	Different platforms may present conflicting information about a candidate.	Apply data validation and cross-platform verification.
Privacy Concerns	Handling personal information could raise privacy and ethical issues.	Follow data protection regulations and anonymize sensitive data.

11.4.3 Ethical and Legal Risks

AI-based hiring systems must address potential ethical and legal issues.

Table 7:Ethical and Legal Risks Table

Risk	Description	Mitigation Strategy
Algorithm Bias	AI models may unintentionally favor certain profiles or backgrounds.	Train models using diverse datasets and monitor fairness metrics.
Transparency Issues	Candidates may not understand how hiring decisions are made.	Implement Explainable AI techniques to justify predictions.
Legal Compliance	Recruitment systems must comply with labor laws and data regulations.	Ensure compliance with data protection standards and recruitment policies.